BALA 302 – FINAL ASSIGNMENT

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A Classification-Based Risk Profiling of Drug-Related Ambulance Attendances in Australia: Demographic, Substance-Type and Jurisdictional Determinants in the Post-Pandemic Context

Executive Summary

Drug-related ambulance attendances in Australia present a complex and uneven demand across jurisdictions, demographic groups, and substance categories. Although national surveillance systems provide descriptive reporting, these patterns are not consistently converted into operationally useful risk profiles. This project applies a classification-based analytics framework to identify differentiated risk levels and examine how combinations of demographic, substance-type, and jurisdictional factors are associated with higher-risk attendance profiles across regions in the evolving post-pandemic service environment.

The study uses secondary data from the National Ambulance Surveillance System (NASS), which contains standardized records of drug-related ambulance attendances across Australian jurisdictions. Age group, gender, drug type, jurisdiction, and attendance rate indicators are among the variables included in the analysis. Data were cleaned, structured, and transformed to support classification modelling, with a

binary risk label derived from attendance rate distributions to distinguish higher-risk and lower-risk profiles.

A supervised classification approach was implemented using a train–test split to ensure generalisability. Multiple models were evaluated using accuracy, precision, recall, F1-score, and ROC-AUC. The final model employed a tree-based ensemble method, XG Boost, to capture non-linear relationships and interaction effects between demographic and substance-related variables. An input-based prediction interface was developed to test different demographic, drug, and jurisdictional combinations, enabling scenario-based risk classification. Model interpretation techniques were integrated at both global and individual levels to explain how specific variable combinations contribute to elevated risk profiles across jurisdictions.

The analysis indicates that risk patterns are shaped by the interaction of demographic characteristics, drug categories, and regional contexts rather than single variables in isolation. The ensuing classification outputs strengthen analytical and managerial implications and contribute to data-informed operational decision-making in a changing service landscape by offering structured insights that may support situational awareness, evidence-based resource planning, and focused intervention strategies for ambulance services.

1. Introduction

1.1. Background

Ambulance services in Australia respond to a significant number of alcohol and other drug-related emergencies, representing a considerable share of acute healthcare demand. According to national surveillance data, these attendances differ significantly amongst drug types, demographic groups, and jurisdictions, suggesting irregular emergency care use patterns rather than consistent national trends (Australian Institute of Health and Welfare [AIHW] 2024). The post-pandemic period (2021–2023) has seen evolving patterns in substance use and emergency service demand, including increased stimulant-related attendances and mental health co-

occurring incidents in several jurisdictions (AIHW 2024), motivating a focused examination of risk profiles in this transitional phase.

Systems like the National Ambulance Surveillance System (NASS) offer structured data that facilitates cross-jurisdictional monitoring of ambulance attendances and enhances visibility of substance-related harms (Dietze et al. 2019). Majority of current reporting is still descriptive and does not convert these differences into organised risk profiles that might aid in resource allocation and operational planning.

The central research question guiding this study is:

How can drug-related ambulance attendances in Australia be systematically classified into differentiated risk levels using demographic, substance-type, and jurisdictional characteristics and what factor combinations contribute to elevated risk levels across cases in post pandemic context?

1.2. Problem Statement

Although national surveillance systems provide extensive reporting on drug-related ambulance attendances, existing analyses largely remain descriptive, focusing on aggregate trends rather than predictive risk identification. As a result, it is difficult to rigorously determine which combinations of demographics, drugs, and jurisdictions are most closely linked to an increased risk of ambulance attendance.

This study addresses this gap by applying machine learning classification techniques to develop a structured risk profiling framework that can differentiate high-risk and low-risk ambulance attendance patterns based on demographic, substance-related, and geographic characteristics.

1.3. Business Objective

This framework aims to support operational decision-making by enabling ambulance services to better anticipate demand patterns and prioritise targeted prevention or response strategies

1.4. Research Motivation

Academic

This study adds to the field of literature by showing how explainable AI approaches (SHAP) and machine learning classification methods can be used to discover distinct risk patterns in ambulance monitoring data.

Practical

Practically speaking, the research offers a data-driven methodology for identifying high-risk attendance patterns, which could facilitate more effective emergency response planning, targeted preventative measures, and better resource allocation.

2. Literature Review

Theoretical Background

Risk classification in prehospital and emergency care.

To help services prioritise response, allocate resources, and enhance results, risk stratification divides cases into clinically significant levels (such as low vs. high risk). Classification models are an appropriate analytical framework for structured incident-level data because the term "risk" in prehospital contexts typically refers to severity, likelihood of worsening, reporting, or downstream service demand (Miles et al. 2020; Almulihi et al. 2024).

Population-level differences and health inequalities.

Ambulance use and outcomes are not uniform across populations; demographic and contextual patterns (age, sex, geography, socioeconomic environment) are repeatedly linked to differences in emergency care needs and service use. This supports the

inclusion of jurisdiction/geography and demography as essential factors in the definition of differential risk profiles (Bell et al. 2024; AIHW 2023).

Surveillance datasets as evidence for acute harms and service burden.

Ambulance clinical records are increasingly used for public-health surveillance because they capture acute events and can reveal demographic and temporal patterns not fully visible in hospital-only datasets. Systems like Australia's National Ambulance Surveillance System (NASS) show how coded clinical data with clinical, demographic, and time-based information can be used to analyse ambulance attendances (Lubman et al. 2020).

Review of Prior Studies

A) Studies with Ambulance and prehospital Emergency data

Demographic patterning in ambulance attendances.
Australian reporting shows strong differences by age and sex in specific ambulance attendance categories (e.g., self-harm/suicidality), reinforcing that demographics can meaningfully differentiate risk-linked attendance profiles (AIHW 2023; Lubman et al. 2020).

Ambulance datasets as structured, coded clinical records (Australia).
NASS is described as a national, population-level surveillance system derived from paramedic clinical records with extensive variables including demographics, temporal and geospatial factors, and clinical indicators—supporting feasibility of structured modelling for risk differentiation (Lubman et al. 2020).

Jurisdiction/geography and large-scale attendance patterns.
Large linked analyses of older adult ambulance attendances (multi-year, high volume) show that attendances vary by age bands, sex, and urban versus non-urban location, indicating that geographic and jurisdictional factors are associated with systematic differences in ambulance service use patterns (Xu et al. 2025; Bell et al. 2024).

Due to its interpretability, logistic regression was frequently used in early emergency risk prediction models. However, these methods may have trouble capturing intricate non-linear relationships in structured healthcare data (Steyerberg et al. 2010). To enhance prediction accuracy, more recent research has been using machine learning techniques, especially tree-based ensembles; nonetheless, these models present interpretability issues in operational and clinical contexts (Rudin, 2019).

B) Tree based models in structured classification problems

Margin-based classification using Support Vector Machines (SVM).
Supervised learning models called Support Vector Machines are made to find the best separation hyperplane that maximises the margin between classes. Support Vector Machines identify an optimal separating hyperplane and can model non-linear relationships using kernel functions (Cortes & Vapnik 1995).

Gradient-boosted tree models (XG Boost) for structured data.

Gradient boosting methods construct sequential decision trees in which each new tree corrects the errors of the previous ensemble. XG Boost is a scalable implementation of gradient-boosted trees designed for efficiency, regularization, and high predictive accuracy in structured/tabular datasets (Chen & Guestrin 2016).

C) Explainability in Machine Learning Models

Model Interpretability in Healthcare Analytics
The increasing use of complex machine learning models in healthcare has raised concerns regarding transparency, interpretability, and trust. Black-box models may achieve high predictive accuracy but often lack explainability, limiting their applicability in operational decision-making environments (Carriero et al. 2025; Mienye et al. 2024). Explainability is particularly important in healthcare contexts where model outputs may influence clinical or operational decisions, requiring interpretable reasoning rather than purely predictive performance (Rudin 2019).

SHAP (Shapley's Additive Explanations) for feature attribution

Recent empirical studies and methodological analyses indicate that SHAP is particularly well-suited for explaining gradient-boosted tree models such as XG Boost because TreeSHAP leverages the internal tree structure to compute exact Shapley values efficiently and consistently (Lundberg et al. 2020). SHAP maintains local accuracy and consistency guarantees in contrast to model-agnostic explanation strategies, guaranteeing that feature attributions correspond with variations in model output (Lundberg & Lee, 2017).

In structured tabular datasets—where XG Boost is frequently reported to outperform many alternative algorithms—SHAP enables transparent identification of global importance patterns while also explaining individual risk classifications at the instance level (Chen & Guestrin 2016; Rudin 2019). SHAP is one of the most theoretically sound and frequently used interpretability frameworks in tree-based clinical and operational prediction models, according to systematic evaluations of explainable AI in healthcare (Mienye et al. 2024; Carriero et al. 2025).

XG Boost's built-in feature importance metrics (e.g., gain or split frequency) summarize how often and how effectively variables are used in tree splits but do not provide consistent, instance-level explanations and may be biased toward high-cardinality features (Chen & Guestrin, 2016). In contrast, SHAP assigns contributions based on Shapley values, ensuring local accuracy and consistency, while enabling both global and individual-level interpretation for tree ensembles (Lundberg & Lee 2017; Lundberg et al. 2020).

Many healthcare machine learning studies report global feature importance to summarise overall variable influence; however, such aggregate measures do not explain individual predictions. Instance-level explanations are necessary to understand case-specific risk classification in operational settings (Ribeiro et al. 2016; Lundberg & Lee 2017).

Analytical Implications and Gap

Prior research establishes demographic and jurisdictional variation in ambulance attendances and demonstrates increasing use of machine learning for emergency risk prediction. However, much of the literature prioritises predictive performance and reports aggregate feature importance without systematically integrating instance-level explainability frameworks. Limited emphasis has been placed on combining structured classification with transparent, case-level interpretation in ambulance datasets. Accordingly, this study integrates supervised classification with SHAP-based explanation to identify differentiated risk profiles across demographic and jurisdictional characteristics, while ensuring transparency and robustness in model evaluation.

3. Methodology

3.1. Analytical Approach

This study adopts a supervised machine learning classification framework to predict high-risk ambulance attendance profiles. The objective is to classify observations into two categories — High Risk and Low Risk — based on demographic, jurisdictional, and substance-related variables.

A classification approach was selected instead of regression because the research objective focuses on risk stratification rather than continuous rate forecasting. In public health contexts, predictive models are typically used to support triage and prioritisation by identifying high-risk groups rather than estimating precise continuous outcomes (Wynants et al. 2020). Prioritisation, focused intervention, and organised decision-making are all aided by risk classification.

Several modelling approaches were used and contrasted to guarantee methodological integrity. Both linear and non-linear analytical cases are represented by the models:

- Logistic Regression
- Random Forest
- Gradient Boosting
- Extreme Gradient Boosting (XG Boost)
- Support Vector Machine (SVM)

The inclusion of diverse modelling approaches allows proper evaluation of whether risk patterns are primarily linear or whether complex non-linear interactions significantly improve prediction performance.

Logistic Regression was included as a baseline because it is widely used in healthcare risk modelling due to interpretability and statistical transparency (Hosmer, Lemeshow & Sturdivant 2013). However, previous studies show that when interaction effects and non-linear correlations are present in structured healthcare datasets, ensemble tree-based approaches often perform better than linear models (Chen & Guestrin 2016).

SVM was included because margin-based classifiers are effective in high-dimensional spaces and provide strong separation performance (Cortes & Vapnik 1995).

According to Chen and Guestrin (2016), XG Boost was chosen because of its resilience, and excellent performance in tabular data contexts.

Strong analytical reasoning is guaranteed by this comparison modelling design, which also lessens model-selection bias.

3.2. Data Preparation

The ambulance attendance rate per 100,000 people was a continuous metric in the dataset. This variable was utilised to create a binary result that matched the study's goal of risk classification. This transformation enabled systematic comparison of classification algorithms by defining a clear binary outcome variable representing high-risk and low-risk attendance profiles.

The attendance rate was eliminated from the feature set once the binary result was constructed. Since the risk classification is directly determined by the attendance rate, this step was required to prevent target leakage. By adding it as a predictor, the model would have indirect access to the outcome variable, which would compromise validity and inflate performance (Kaufman et al. 2012). Instead of mechanically replicating the result, by eliminating it, the model finds separate demographic and drug-related determinants of heightened risk.

3.2.1. Decision Threshold for Risk Classification

The median attendance rate was used as the decision threshold to construct the binary outcome variable. Observations above the median were classified as High Risk, while those below were classified as Low Risk. The median was selected because it offers a neutral, data-driven cutoff that prevents severe skew in class distribution and guarantees fair representation of risk categories during model training. The median provides a data-driven cutoff that maintains balanced class distribution when predefined clinical thresholds are unavailable, making it suitable for risk stratification in healthcare analytics (Steyerberg et al. 2010).

3.2.2. Selection of Independent Variables

The independent variables included Age Group, Gender, Drug Type, State, and Year. These variables were selected based on prior literature identifying demographic factors and substance categories as key drivers of ambulance attendance and health service demand, aligning with the study's aim to identify structural risk factors. Only pre-outcome variables were included to ensure that the model captures independent explanatory factors rather than post-event indicators.

3.2.3. Missing Value Treatment

Imputation was used to manage missing values to prevent the bias that comes with listwise deletion. The most frequent category was used to impute categorical data,

whilst the median was used to impute numerical variables. Median imputation is commonly used in healthcare datasets due to its robustness to skew distributions and outliers (Little & Rubin 2019). This approach preserves sample size while maintaining distributional stability.

3.2.4. Encoding of Categorical Variables

One-hot encoding was applied so that categorical variables could be used correctly by all models without assuming any order between categories. One-hot encoding is widely recommended for structured tabular data in machine learning applications (James et al. 2021). To preserve model stability, the encoding procedure was set up to disregard unknown categories during testing.

3.2.5. Feature Scaling

Feature scaling was applied selectively. Standardisation (StandardScaler) was used for the Support Vector Machine model, as margin-based classifiers are sensitive to feature magnitude (Cortes & Vapnik 1995). On the other hand, because tree-based models like Random Forest, Gradient Boosting, and XG Boost are invariant to monotonous changes in input variables, they were trained without scaling. This guarantees proper and equitable methodological comparisons between model types.

3.2.6. Train – Test Split

The dataset was partitioned into 80% training data and 20% testing data. Stratified sampling was used to maintain the distribution of classes across splits. To assure repeatability, a fixed random seed was used. To avoid data leaks and guarantee uniform transformation across the training and testing stages, all preprocessing procedures were implemented within scikit-learn pipelines.

3.3. Model Implementation and Evaluation Framework

All models were implemented under identical preprocessing pipelines to ensure fair comparison and prevent data leakage. Default hyperparameters were used to maintain methodological consistency.

Performance was evaluated using Accuracy, Precision, Recall, F1-score, and ROC-AUC. F1-score measures the ratio of false positives to false negatives, whereas accuracy indicates total correctness. Because it evaluates discrimination across all classification thresholds and is less subject to threshold selection, ROC-AUC was chosen as the major statistic (Steyerberg et al. 2010).

3.4. Model Explainability

The final XG Boost model was subjected to SHAP (SHapley Additive Explanations) to guarantee interpretability. By quantifying each feature's contribution to the expected risk outcome, SHAP makes it possible to clearly explain how the model behaves.

Global SHAP analysis was conducted to identify overall feature importance and the direction of influence on High-Risk classification. In addition, input-based prediction testing was performed by providing specific demographic and drug-type combinations to the trained model. For each case, local SHAP waterfall plots were generated to explain how individual features increased or decreased the predicted probability of High Risk.

This approach ensures that the final model is not only accurate but also interpretable at both the aggregate and individual case levels.

4. Analysis and Results

4.1. Descriptive Analysis

Descriptive statistics were first examined to understand the overall structure and distribution of the ambulance attendance dataset before model development.

About 1,440 observations, representing drug-related ambulance attendances across various age groups, genders, drug categories, jurisdictions, and years (2021–2023), are included in the dataset. The attendance rate variable represents the number of ambulance incidents per 100,000 population.

The mean attendance rate is approximately 163.67 per 100,000 population, while the median is 28.9, indicating that the distribution is highly right-skewed. The maximum value of 4954 and the significant difference between the mean and median point to the existence of extremely high-attendance cases in specific drug-demographic-jurisdiction combinations.

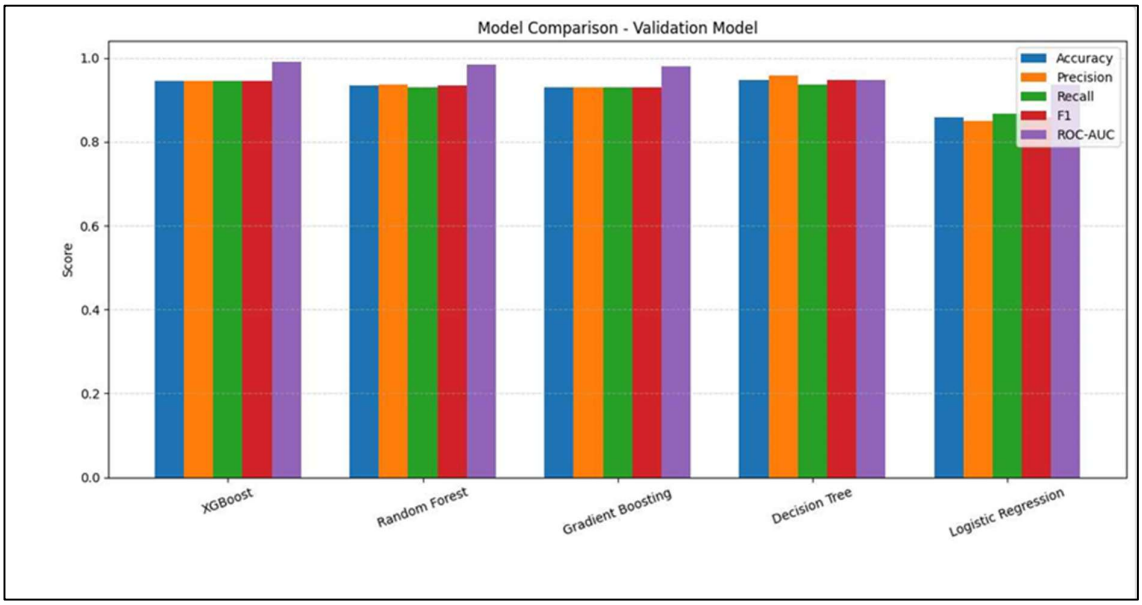
Additionally, categorical variables exhibit distinct patterns of concentration. The most common drug type is cannabis (frequency = 180), the most common gender group is female (frequency = 720), and the most common jurisdiction is New South Wales (NSW) (frequency = 240). These trends imply that the distribution of ambulance attendances across regional and demographic groups is not uniform.

The dataset is roughly balanced between high-risk and low-risk observations, according to the binary risk variable derived from the median attendance rate, which has a mean value of 0.498. This balanced distribution lessens the possibility of class imbalance bias during model training and supports the appropriateness of classification modelling.

Overall, the descriptive analysis supports the need for a classification framework to identify distinct risk profiles by confirming that ambulance attendance patterns range significantly across demographic features, drug kinds, and jurisdictions.

4.2. Model Building / Testing

To assess predictive performance, multiple classification algorithms were evaluated using discrimination metrics. The Model Comparison Graph summarises the performance of each model across these evaluation metrics.



Model Comparison Graph

The comparison shows that tree-based ensemble models consistently outperform linear models. Strong classification capability is demonstrated by the high values of XG Boost, Random Forest, Gradient Boosting, and Decision Tree across all evaluation metrics. Logistic regression, on the other hand, performs substantially worse, especially in ROC-AUC, indicating that the intricacy of ambulance attendance risk patterns cannot be adequately captured by simple linear correlations.

According to the preliminary model comparison results, Support Vector Machine (SVM) appeared to achieve the highest ROC–AUC, indicating strong discrimination between high-risk and low-risk observations. XG Boost followed closely behind, while Random Forest and Gradient Boosting also demonstrated high predictive accuracy.

However, further validation was carried out because the goal was not only to maximise predicted performance but also to guarantee model stability and interpretability. To confirm their resilience, the two best-performing algorithms—SVM

and XG Boost—were reassessed using their whole end-to-end consistently pre-processed pipelines and final hyper parameter tuning.

	XG Boost	**SVM**
Accuracy	0.94	0.844
ROC- AUC	0.991	0.932

Performance Comparison of Top Performing Models

The additional testing revealed that XG Boost consistently outperformed SVM, achieving both higher accuracy and a stronger ROC–AUC score. The model is quite successful in differentiating between high-risk and low-risk ambulance attendance profiles across various probability thresholds, as indicated by the ROC–AUC value of 0.991

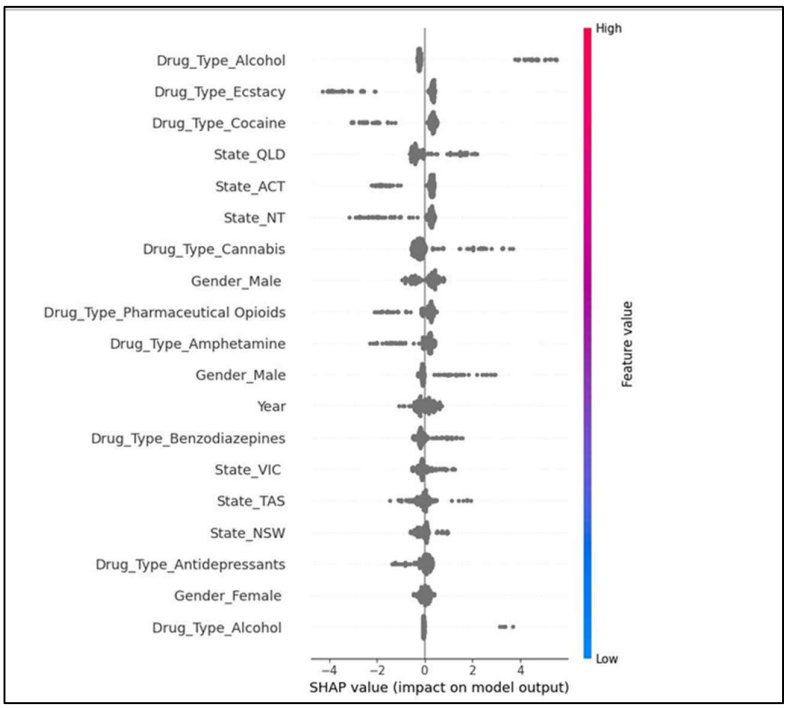
The small difference between the initial model comparison and the pipeline-level evaluation can occur due to randomness in model training. Ensemble models such as boosting methods rely on random sampling during training (Breiman 2001; Chen & Guestrin 2016). In addition, Support Vector Machines are sensitive to feature scaling and margin optimisation, which can influence classification boundaries under different preprocessing settings (Cortes & Vapnik 1995).

Overall, the analysis indicates that tree-based ensemble models are better suited for this dataset because they can capture complex interactions between demographic characteristics, drug categories, and jurisdictional factors. Among the evaluated algorithms, XG Boost demonstrated the strongest and most consistent predictive performance. Its ability to model non-linear relationships and interaction effects makes it particularly effective for structured healthcare data (Chen & Guestrin 2016). In addition, its compatibility with SHAP-based explainability techniques enables transparent interpretation of the factors contributing to high-risk classifications.

Therefore, XG Boost was selected as the final model for risk stratification, providing the best balance between predictive accuracy, robustness, and interpretability.

4.3. Key Findings

To identify the variables driving high-risk predictions, SHAP analysis was applied to the final XG Boost model. The global SHAP summary plot highlights the variables that contribute most strongly to risk classification.



Global SHAP Feature Importance for High-Risk Classification.

The results indicate that drug type is the strongest predictor of ambulance attendance risk, followed by age group and jurisdiction. Strong contributions to high-risk predictions are indicated by the huge positive SHAP values of some drug categories, including alcohol, cannabis, cocaine, and ecstasy. Among the most significant predictors are age groups like 15–24 and 55+. In addition, jurisdictional variables including Queensland, Northern Territory, and ACT demonstrate notable influence, suggesting that ambulance attendance risk varies across regional contexts.

To further interpret the behaviour of the model, an input-based prediction interface was used to test specific demographic and drug-related combinations. This interface allows the trained model to generate predicted risk outcomes for selected scenarios.

```
Enter input values:

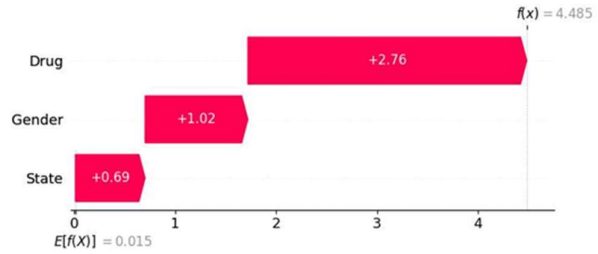
Age_Group (e.g., 15-24 / 25-34 / 35-44): 15-24
Gender (e.g., Male / Female): Male
Drug_Type (e.g., Cannabis / Alcohol / Amphetamines / Ecstasy): Cannabis
State (e.g., NSW / QLD / NT / TAS / VIC / ACT): NSW
Year (e.g., 2021): 2021

==============================
PREDICTION RESULT
==============================
Prediction: HIGH RISK
High-risk probability: 0.993
Low-risk probability : 0.007

==============================
WHY THE MODEL PREDICTED THIS
==============================
Main drivers pushing toward HIGH risk:
- Drug increased risk (SHAP=+2.762)
- Gender increased risk (SHAP=+1.018)
- State increased risk (SHAP=+0.690)

Drivers reducing risk (but not enough to flip):
- Year reduced risk (SHAP=-0.051)

--- SHAP Waterfall (HIGH risk drivers only) ---
```




```
(1,
 0.9929551482200623,
   orig_col      shap       abs
0      Drug  2.761992  2.761992
1    Gender  1.017840  1.017840
2     State  0.690229  0.690229
3      Year -0.051319  0.051319)
```

For the input combination: Age Group 15–24, Male, Cannabis use, and NSW (2021), the model predicted a High-Risk probability of 0.993. The SHAP waterfall plot shows that drug type contributed the most to the risk (+2.76), followed by gender (+1.02) and state (+0.69). Although the year slightly reduced the risk, the combined influence of the other variables pushed the prediction toward a High-Risk classification.

```
Enter input values:

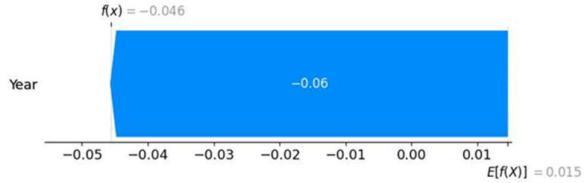
Age_Group (e.g., 15-24 / 25-34 / 35-44): 25-34
Gender (e.g., Male / Female): Male
Drug_Type (e.g., Cannabis / Alcohol / Amphetamines / Ecstasy): Ecstacy
State (e.g., NSW / QLD / NT / TAS / VIC / ACT): ACT
Year (e.g., 2021): 2022

==============================
PREDICTION RESULT
==============================
Prediction: LOW RISK
High-risk probability: 0.160
Low-risk probability : 0.840

==============================
WHY THE MODEL PREDICTED THIS
==============================
Main drivers keeping this case LOW risk:
- Year reduced risk (SHAP=-0.060)

Drivers increasing risk slightly (but not enough to flip):
- Drug slightly increased risk (SHAP=+2.727)
- Gender slightly increased risk (SHAP=+1.147)
- State slightly increased risk (SHAP=+0.900)

--- SHAP Waterfall (LOW risk drivers only) ---
```




```
(0,
 0.15983061492443085,
   orig_col      shap       abs
0     Drug  2.726831  2.726831
1   Gender  1.147321  1.147321
2    State  0.899906  0.899906
3     Year -0.060167  0.060167)
```

For the input combination: Age Group 25–34, Male, Ecstasy use, ACT (2022), the model predicted Low Risk (0.16), mainly because the year variable reduced the overall risk, offsetting the smaller risk contributions from the other factors.

Overall, the findings show that rather than being influenced by single variables, the probability of ambulance attendance is influenced by interactions between medication type, demographic traits, and jurisdictional considerations. These non-linear correlations were effectively captured by the XG Boost model, allowing for precise distinction between high-risk and low-risk profiles. Further confirmation of how particular variable combinations influence risk estimates was provided by SHAP analysis, which offers clear insights into the elements influencing increased ambulance attendance risk.

5. Discussion

Theoretical Contributions

This study contributes to the growing intersection between public health surveillance and explainable machine learning by demonstrating how ambulance attendance data can be operationalized into risk-stratified profiles rather than descriptive statistics

alone. This study expands theoretical work on explainable health analytics by combining classification modelling with SHAP-based interpretability, demonstrating how combinations of drug types, jurisdictions, and demographic characteristics interact to affect emergency service demand (Lundberg & Lee 2017).

Managerial Contributions

The model gives emergency service planners a useful foundation for identifying drug-type combinations and high-risk demographics across jurisdictions, enabling better operational planning and earlier intervention. For example, regions where stimulant-related incidents coincide with younger populations may require additional ambulance capacity, localized harm-reduction programs, and community outreach initiatives. By translating surveillance data into actionable risk insights, the framework supports evidence-based emergency planning, improved coordination between health authorities, and more efficient deployment of ambulance services in areas facing elevated drug-related harm.

6. Recommendations

- Incorporate SHAP-based risk alerts into ambulance dispatch dashboards to allow planners to track jurisdiction-level risk signals in real time and predict times when demand for drugs will be higher. Dispatch centres can increase reaction readiness and deploy units more effectively by integrating explainable AI results into operational dashboards (Lundberg & Lee 2017; Rajkomar et al. 2019). (OPERATIONAL)
- Link ambulance surveillance data with hospital emergency department admissions to improve short- to medium-term monitoring of drug-related harm. Cross-agency integration would enable earlier detection of emerging substance trends and escalation patterns that are not visible within ambulance data alone. (TACTICAL)

- Create a national drug-risk forecasting system based on data from ambulance surveillance. Health authorities might monitor new substance trends, model future demand scenarios, and coordinate national emergency preparedness measures by aggregating predictive outputs across states (Rajkomar et al. 2019; AIHW 2024). (LONG TERM STRATEGY)

7. Limitations

- The study relies on surveillance data from 2021–2023, which may not fully capture long-term shifts in drug markets, substance availability, or behavioural patterns influencing ambulance attendances. Drug trends can evolve rapidly, particularly with emerging synthetic substances and changing consumption environments.
- The dataset only includes incidents that resulted in ambulance attendance, meaning drug-related harms that did not require emergency services are not captured. Therefore, rather than reflecting the entire existence of drug use or harm in the population, surveillance databases reflect acute service demand (Dietze et al. 2019; AIHW 2024).
- Predictive models' explanatory depth may be limited by the dataset's lack of significant contextual drivers such socioeconomic conditions, mental health status, and environmental risk factors. Research indicates that more general social variables have a substantial impact on health outcomes connected to substance use (Marmot et al. 2020).

Appendix

```python
from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_score, roc_auc_score

def get_pos_score(model, X_eval):
    """
    Returns a score for ROC-AUC:
    - predict_proba[:, class1] if available
    - else decision_function if available
    """
    if hasattr(model, "predict_proba"):
        proba = model.predict_proba(X_eval)
        if hasattr(model, "classes_"):
            idx1 = int(np.where(model.classes_ == 1)[0][0])
            return proba[:, idx1]
        return proba[:, 1]
    if hasattr(model, "decision_function"):
        return model.decision_function(X_eval)
    return None

rows = []

for name, model in models_val.items():
    # Use cleaned X for SVM pipelines, normal X_test for others (safe default)
    X_eval = X_test_clean if "SVM" in name else X_test

    y_pred = model.predict(X_eval)
    score = get_pos_score(model, X_eval)

    auc = roc_auc_score(y_test, score) if score is not None else np.nan

    rows.append({
        "Model": name,
        "Accuracy": accuracy_score(y_test, y_pred),
        "Precision": precision_score(y_test, y_pred, zero_division=0),
        "Recall": recall_score(y_test, y_pred, zero_division=0),
        "F1": f1_score(y_test, y_pred, zero_division=0),
        "ROC-AUC": auc
    })

compare_df = pd.DataFrame(rows).sort_values("ROC-AUC", ascending=False).reset_index(drop=True)
compare_df
```

	Model	Accuracy	Precision	Recall	F1	ROC-AUC
0	SVM (Linear)	0.968858	0.985612	0.951389	0.968198	0.996648
1	XGBoost	0.944837	0.944444	0.944444	0.944444	0.990469
2	SVM (RBF)	0.941176	0.963504	0.916667	0.939502	0.987598
3	Random Forest	0.934256	0.937063	0.930556	0.933798	0.985034
4	Gradient Boosting	0.930796	0.930556	0.930556	0.930556	0.980651
5	Decision Tree	0.948097	0.957447	0.937500	0.947368	0.948060
6	Logistic Regression	0.858131	0.850340	0.868056	0.859107	0.936590

Machine Learning Model Performance Comparison

```python
import matplotlib.pyplot as plt
import numpy as np

metrics = ["Accuracy", "Precision", "Recall", "F1", "ROC-AUC"]
plot_df = compare_df.set_index("Model")[metrics]

x = np.arange(len(plot_df.index))
bar_w = 0.14

plt.figure(figsize=(13, 5))
for i, m in enumerate(metrics):
    plt.bar(x + i*bar_w, plot_df[m].values, width=bar_w, label=m)

plt.xticks(x + (len(metrics)-1)*bar_w/2, plot_df.index, rotation=20, ha="right")
plt.ylim(0, 1.05)
plt.title("Model Comparison (Test Set)")
plt.ylabel("Score")
plt.grid(axis="y", linestyle="--", alpha=0.4)
plt.legend()
plt.tight_layout()
plt.show()
```

Model Comparison Graph code

```python
# 1) Imports
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.compose import ColumnTransformer
from sklearn.pipeline import Pipeline
from sklearn.preprocessing import OneHotEncoder
from sklearn.impute import SimpleImputer

from sklearn.metrics import (
    accuracy_score, confusion_matrix, classification_report,
    roc_auc_score, roc_curve, precision_score, recall_score, f1_score
)

from xgboost import XGBClassifier

import shap
```

Library Imports and Environment Setup

```python
print("\nDescriptive Statistics:")
print(df.describe(include='all'))
```

```
Descriptive Statistics:
       Attendance_Rate (per 100,000  population)   Gender Drug_Type State  \
count                                1439.000000     1440      1440  1440
unique                                       NaN        3         9     6
top                                          NaN   Female   Cannabis   NSW
freq                                         NaN      720       180   240
mean                                  163.674844      NaN       NaN   NaN
std                                   451.502939      NaN       NaN   NaN
min                                     0.000000      NaN       NaN   NaN
25%                                     0.000000      NaN       NaN   NaN
50%                                    28.900000      NaN       NaN   NaN
75%                                   101.950000      NaN       NaN   NaN
max                                  4954.000000      NaN       NaN   NaN

              Year  High_Risk_Attendance
count   1440.00000           1441.000000
unique         NaN                   NaN
top            NaN                   NaN
freq           NaN                   NaN
mean    2022.00000              0.498265
std        0.81678              0.500171
min     2021.00000              0.000000
25%     2021.00000              0.000000
50%     2022.00000              0.000000
75%     2023.00000              1.000000
max     2023.00000              1.000000
```

```python
# 2) Load data
CSV_PATH = "BALA_302_FINAL_COMPLETE.csv"

df = pd.read_csv(CSV_PATH)
df = df.loc[:, ~df.columns.str.contains(r"^Unnamed", na=False)]

print("Shape:", df.shape)
print(df.head())
print("\nColumns:\n", df.columns.tolist())
```

```
Shape: (1441, 7)
        Gender Drug_Type State    Year  \
0  15-24   Male   Alcohol   NSW  2021.0
1  15-24   Male  Alcohol   VIC   2021.0
2  15-24   Male   Alcohol   QLD  2021.0
3  15-24   Male   Alcohol   TAS  2021.0
4  15-24  Male   Alcohol    ACT  2021.0

   Attendance_Rate (per 100,000  population)  High Risk Attendance
0                                      569.8                     1
1                                      654.1                     1
2                                      955.0                     1
3                                      959.3                     1
4                                      852.9                     1

Columns:
 [' ', 'Gender', 'Drug_Type', 'State', 'Year', 'Attendance_Rate (per 100,000  population)', 'High Risk Attendance ']
```

Dataset Loading and Descriptive Analysis

```python
# 3) Set column names (EDIT THESE)
# Attendance column (used only to build/confirm the target)
COL_ATT = "Attendance_Rate (per 100,000  population)"
COL_Y = "High_Risk_Attendance"

CAT_FEATURES = ["Age_Group", "Gender", "Drug_Type", "State"]
NUM_FEATURES = ["Year"]

keep_cols = [COL_ATT] \
    + [c for c in CAT_FEATURES if c in df.columns] \
    + [c for c in NUM_FEATURES if c in df.columns] \
    + ([COL_Y] if COL_Y in df.columns else [])

df = df[keep_cols].copy()
print("\nUsing columns:", df.columns.tolist())
```

```
Using columns: ['Attendance_Rate (per 100,000  population)', 'Gender', 'Drug_Type', 'State', 'Year']
```

```python
# 4) Create/confirm target from median attendance
median_att = df[COL_ATT].median()
df[COL_Y] = (df[COL_ATT] > median_att).astype(int)

print("Median attendance:", median_att)
print("Target counts:\n", df[COL_Y].value_counts())
```

```
Median attendance: 28.9
Target counts:
 High_Risk_Attendance
0    723
1    718
Name: count, dtype: int64
```

```
#5) Build X/y (REMOVE attendance from X)
X = df.drop(columns=[COL_ATT, COL_Y]).copy()
y = df[COL_Y].astype(int)

cat_cols = X.select_dtypes(include=["object"]).columns.tolist()
num_cols = X.select_dtypes(include=["number"]).columns.tolist()

print("X columns:", X.columns.tolist())
print("Categorical:", cat_cols)
print("Numeric:", num_cols)
```

```
X columns: ['Gender', 'Drug_Type', 'State', 'Year']
Categorical: ['Gender', 'Drug_Type', 'State']
Numeric: ['Year']
```

Feature Engineering and Target Variable Definition

```
# 6) Train/test split
X_train, X_test, y_train, y_test = train_test_split(
    X, y,
    test_size=0.2,
    random_state=42,
    stratify=y
)

print("Train:", X_train.shape, "Test:", X_test.shape)
```

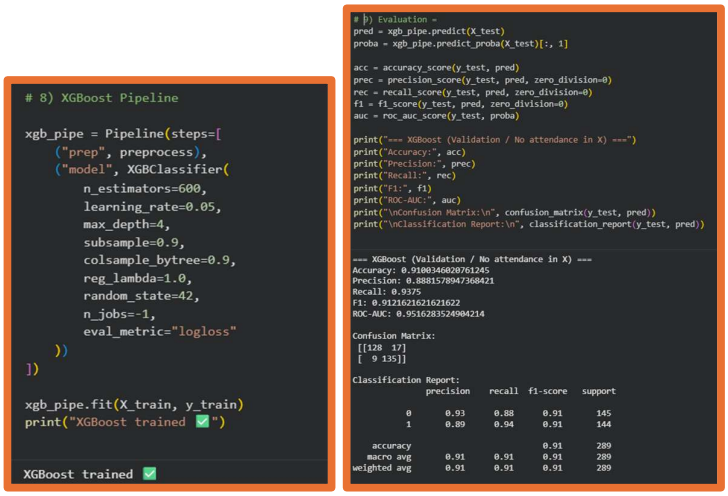
```
Train: (1152, 4) Test: (289, 4)
```

Training and Testing Data Partition

```
# 7) Preprocess (Impute + OneHot)
preprocess = ColumnTransformer(
    transformers=[
        ("num", Pipeline(steps=[
            ("imputer", SimpleImputer(strategy="median"))
        ]), num_cols),

        ("cat", Pipeline(steps=[
            ("imputer", SimpleImputer(strategy="most_frequent")),
            ("onehot", OneHotEncoder(handle_unknown="ignore"))
        ]), cat_cols),
    ]
)
```

Data Preprocessing Pipeline



```
# 8) XGBoost Pipeline

xgb_pipe = Pipeline(steps=[
    ("prep", preprocess),
    ("model", XGBClassifier(
        n_estimators=600,
        learning_rate=0.05,
        max_depth=4,
        subsample=0.9,
        colsample_bytree=0.9,
        reg_lambda=1.0,
        random_state=42,
        n_jobs=-1,
        eval_metric="logloss"
    ))
])

xgb_pipe.fit(X_train, y_train)
print("XGBoost trained ✅")

XGBoost trained ✅
```

```
# 9) Evaluation
pred = xgb_pipe.predict(X_test)
proba = xgb_pipe.predict_proba(X_test)[:, 1]

acc = accuracy_score(y_test, pred)
prec = precision_score(y_test, pred, zero_division=0)
rec = recall_score(y_test, pred, zero_division=0)
f1 = f1_score(y_test, pred, zero_division=0)
auc = roc_auc_score(y_test, proba)

print("=== XGBoost (Validation / No attendance in X) ===")
print("Accuracy:", acc)
print("Precision:", prec)
print("Recall:", rec)
print("F1:", f1)
print("ROC-AUC:", auc)
print("\nConfusion Matrix:\n", confusion_matrix(y_test, pred))
print("\nClassification Report:\n", classification_report(y_test, pred))


=== XGBoost (Validation / No attendance in X) ===
Accuracy: 0.9100346020761245
Precision: 0.8881578947368421
Recall: 0.9375
F1: 0.9121621621621622
ROC-AUC: 0.9516283524904214

Confusion Matrix:
 [[128  17]
 [  9 135]]

Classification Report:
              precision    recall  f1-score   support

           0       0.93      0.88      0.91       145
           1       0.89      0.94      0.91       144

    accuracy                           0.91       289
   macro avg       0.91      0.91      0.91       289
weighted avg       0.91      0.91      0.91       289
```

XG Boost pipeline and Evaluation

```
#  13) SHAP Setup (Global explanations)

X_test_trans = xgb_pipe.named_steps["prep"].transform(X_test)

xgb_model = xgb_pipe.named_steps["model"]

explainer = shap.TreeExplainer(xgb_model)
shap_values = explainer.shap_values(X_test_trans)

shap.summary_plot(
    shap_values,
    features=X_test_trans,
    feature_names=feature_names,
    show=True
)
```

SHAP Global Explanation

```
import numpy as np
import pandas as pd
import shap



xgb_model = xgb_pipe.named_steps["model"]
explainer_margin = shap.TreeExplainer(xgb_model)  ues


def predict_explain_user_input_shap(top_k=5, max_display=10):
    """
    One input model for everyone.
    SHAP waterfall graph is DIFFERENT for HIGH vs LOW by filtering drivers:
      - HIGH: show positive SHAP drivers only
      - LOW : show negative SHAP drivers only
    """


    print("\nEnter input values:\n")
    age   = input("Age_Group (e.g., 15-24 / 25-34 / 35-44): ").strip()
    gen   = input("Gender (e.g., Male / Female): ").strip()
    drug  = input("Drug_Type (e.g., Cannabis / Alcohol / Amphetamines / Ecstasy): ").strip()
    state = input("State (e.g., NSW / QLD / NT / TAS / VIC / ACT): ").strip()
    year  = float(input("Year (e.g., 2021): ").strip())

    input_dict = {"Age_Group": age, "Gender": gen, "Drug_Type": drug, "State": state, "Year": year}
    input_df = pd.DataFrame([input_dict], columns=X.columns)


    pred_label = int(xgb_pipe.predict(input_df)[0])
    p_high = float(xgb_pipe.predict_proba(input_df)[0][1])

    print("\n==============================")
    print("PREDICTION RESULT")
    print("==============================")
    print("Prediction:", "HIGH RISK" if pred_label == 1 else "LOW RISK")
    print(f"High-risk probability: {p_high:.3f}")
    print(f"Low-risk probability : {(1 - p_high):.3f}")
```

```
X_trans = xgb_pipe.named_steps["prep"].transform(input_df)
X_dense = X_trans.toarray() if hasattr(X_trans, "toarray") else X_trans

shap_vals = explainer_margin.shap_values(X_dense)[0]  # 1D array
base_val = float(explainer_margin.expected_value)

shap_df = pd.DataFrame({"feature": feature_names, "shap": shap_vals})

# Map one-hot back to original column names
def to_original(feat):
    if feat in num_cols:
        return feat
    return feat.split("_", 1)[0]

shap_df["orig_col"] = shap_df["feature"].apply(to_original)

# Group to original columns
grouped = shap_df.groupby("orig_col", as_index=False)["shap"].sum()
grouped["abs"] = grouped["shap"].abs()
grouped = grouped.sort_values("abs", ascending=False).reset_index(drop=True)


inc = grouped[grouped["shap"] > 0].copy()
dec = grouped[grouped["shap"] < 0].copy()

print("\n==============================")
print("WHY THE MODEL PREDICTED THIS")
print("==============================")

if pred_label == 1:
    print("Main drivers pushing toward HIGH risk:")
    if inc.empty:
        print("- None (rare)")
    else:
        for _, r in inc.head(top_k).iterrows():
            print(f"- {r['orig_col']} increased risk (SHAP={r['shap']:+.3f})")

    if not dec.empty:
        print("\nDrivers reducing risk (but not enough to flip):")
        for _, r in dec.head(min(3, len(dec))).iterrows():
            print(f"- {r['orig_col']} reduced risk (SHAP={r['shap']:+.3f})")
```

```
    else:
        print("Main drivers keeping this case LOW risk:")
        if dec.empty:
            print("- None (rare)")
        else:
            for _, r in dec.head(top_k).iterrows():
                print(f"- {r['orig_col']} reduced risk (SHAP={r['shap']:+.3f})")

        if not inc.empty:
            print("\nDrivers increasing risk slightly (but not enough to flip):")
            for _, r in inc.head(min(3, len(inc))).iterrows():
                print(f"- {r['orig_col']} slightly increased risk (SHAP={r['shap']:+.3f})")


    if pred_label == 1:

        plot_df = inc.sort_values("shap", ascending=False).head(max_display)
        title = "SHAP Waterfall (HIGH risk drivers only)"
    else:

        plot_df = dec.sort_values("shap", ascending=True).head(max_display)
        title = "SHAP Waterfall (LOW risk drivers only)"


    if plot_df.empty:
        plot_df = grouped.head(max_display)
        title += " [fallback: top absolute drivers]"


    exp = shap.Explanation(
        values=plot_df["shap"].values,
        base_values=base_val,
        data=None,
        feature_names=plot_df["orig_col"].values
    )

    print(f"\n--- {title} ---")
    shap.plots.waterfall(exp, max_display=max_display)

    return pred_label, p_high, grouped
predict_explain_user_input_shap(top_k=5, max_display=10)
```

Input based Prediction model code

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